

Mutations at Novel Sites in *pmrA/B* and *lpxA/D* Genes and Absence of Reduced Fitness in Colistin-Resistant *Acinetobacter baumannii* from a Tertiary Care Hospital, India

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Background: Colistin resistance in *Acinetobacter baumannii*, the last resort drug for serious infections, is emerging worldwide. There has been paucity of data on this aspect from India, which is one of the largest producers of colistin. We studied colistin resistance in *A. baumannii* and characterized the isolates with respect to resistance mechanisms and virulence.

Methods: A total of 365 *A. baumannii* isolates were studied. Antimicrobial susceptibility testing was performed as per standards. Colistin resistance mechanisms were studied by mutation detection in *pmrA/B* and *lpxA/C/D* genes, phenotypic loss of lipopolysaccharide, presence of *mcr1–5* genes, and carbonyl cyanide 3-chlorophenylhydrazide (CCCP) effects. Biofilm formation, desiccation survival, and growth kinetics were studied and statistically analyzed for colistin-resistant and colistin-susceptible isolates.

Results: All the colistin-resistant isolates (9, 2.5%) showed multiple mutations at novel sites in *pmrA/B* and/or *lpxA/D* genes with reversion of resistance with CCCP. Majority of these isolates (6, 66.6%) were from patients without prior colistin therapy. All received prior carbapenems. The resistant isolates demonstrated no significant difference in biofilm formation and desiccation survival but were slow growers.

Conclusion: Mutations in *pmrA/B* and/or *lpxA/D* genes were the main resistance mechanism in these colistin-resistant isolates that showed no reduction in fitness.

Keywords: prior colistin administration, carbapenems, *pmrA* mutation, fitness

Introduction

MULTIDRUG RESISTANCE (MDR) in *Acinetobacter baumannii* has posed a significant challenge for clinicians and researchers in modern health care settings. These well-adapted organisms have been implicated in various hospital-acquired infections, particularly in the intensive care units (ICUs), like infections of respiratory tract, bloodstream, skin, and soft tissue, often in association with invasive devices. It has been enlisted as one of the critical pathogens of priority for research by the World Health Organization (WHO).¹ The easy acquisition and rapid dissemination of MDR in this organism are of particular concern. This has resulted in the use of polymyxins as the last resort therapeutic option for *A. baumannii* infections.

Unfortunately, the intense use of polymyxins, particularly colistin in animal production and farming, has already led to colistin heteroresistance along with emergence of colistin-resistant (Col^R) isolates of *A. baumannii* in few Asian and European countries.^{2–4} The existing carbapenem resistance

in *A. baumannii* has facilitated the easy introduction and rampant usage of colistin in hospital settings. Studies have shown that colistin has also been used before diagnosis in some situations, especially in developing countries like Turkey.⁵ Incidentally, few of the developing countries, including India, are also major producers of colistin in the global market.² Data concerning the burden of colistin resistance in developing countries against *A. baumannii* substantiated by microbiological and molecular characterization of the resistant isolates have been limited, thus failing to reveal the true nature of the impending crisis. Inability to perform colistin susceptibility routinely in majority of the antimicrobial susceptibility testing centers in developing countries might be the reason for scarcity of such reports.

There had been a series of studies on the most probable mechanisms of development of colistin resistance on laboratory-adapted mutants of colistin-susceptible (Col^S) strains.^{6–8} As suggested, colistin resistance develops mostly during the course of colistin therapy.^{9,10} Recently, there have been reports of clinically important colistin-resistant

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A. baumannii across the globe.^{10–12} Colistin resistance in these isolates has been largely attributed to complete loss of lipopolysaccharide (LPS) layer. However, multiple factors contribute to these resistance mechanisms, which are still not very clear.¹⁰ Interestingly, colistin resistance has also been associated with decreased virulence and fitness of the isolates, thus affecting their growth kinetics and non-susceptibility to other antibiotics.¹³

We have previously reported high burden of imipenem-resistant *A. baumannii* amidst intense antibiotic pressure in one of the ICUs of a tertiary care hospital in north India.¹⁴ As high carbapenem resistance often promotes use of colistin, this study was undertaken to characterize the colistin-resistant isolates of *A. baumannii* in a tertiary care hospital, along with elucidation of probable resistance mechanisms and virulence of the resistant isolates.

Materials and Methods

Study site and ethics approval

This prospective study was conducted in the Department of Microbiology, Institute of Medical Sciences, Banaras Hindu University (Varanasi, India), and the associated 1,500-bedded tertiary care hospital, which caters to the need of 5 states of northern and eastern India. The study was approved by the Institute ethics committee (No. Dean/2018/EC/321). Informed consent was taken from the patients for the study.

Bacterial strain

The study was conducted on 9 colistin-resistant isolates out of 356 *A. baumannii* isolates screened over a period of 15 months (April 2017 to June 2018) from various specimens such as endotracheal tube secretions, pus, blood, urine, sputum, cerebrospinal fluid, and other body fluids from patients with clinical diagnosis of infection. For comparison and analysis of virulence factors, nine colistin-susceptible *A. baumannii* isolates (*i.e.*, 1:1 ratio) collected during same time period matching the resistant ones were taken as study controls. All the isolates were identified as *A. baumannii* by multiplex polymerase chain reaction (PCR), targeting *recA* gene and species-specific internal transcribed spacer region gene.¹⁵

Antimicrobial susceptibility testing

The *in vitro* susceptibility pattern of the isolates against the following standard drugs: piperacillin/tazobactam (100/10 µg), ampicillin/sulbactam (10/10 µg), cefepime (30 µg), ceftazidime (30 µg), ceftriaxone (30 µg), meropenem (10 µg), imipenem (10 µg), amikacin (30 µg), gentamicin (10 µg), levofloxacin (5 µg), ciprofloxacin (5 µg), and trimethoprim/sulfamethoxazole (1.25/23.75 µg) was tested by Kirby Bauer disk diffusion method. Furthermore, testing for susceptibility to imipenem, meropenem, doripenem (Sigma-Aldrich Chemicals Pvt. Ltd.), minocycline (GUFIC Biosciences Ltd.), and polymyxin B (Bharat Serums & Vaccines Ltd.), agar dilution method was performed. For colistin (colistin sulfate) and tigecycline (Sigma-Aldrich Chemicals Pvt. Ltd.), broth microdilution method was used as per standard.^{16,17} Testing was performed in triplicates. *Pseudomonas aeruginosa* ATCC[®] 27853, *A. baumannii* ATCC 19606, and *A. baumannii* MTCC[®] 1425 were used as quality controls.

Detection of carbapenemase genes

All the study isolates were screened for the presence of class A (*bla*_{SME}, *bla*_{NMC}, *bla*_{GES}, and *bla*_{KPC}), class B (*bla*_{IMP}, *bla*_{VIM}, and *bla*_{NDM}), and class D (*bla*_{OXA-23}), carbapenemases by multiplex and conventional (uniplex) PCR, respectively, as previously described.^{18,19}

Study of mechanism of colistin resistance

Effect of carbonyl cyanide 3-chlorophenylhydrazone on colistin MIC. Effect of efflux pump inhibitors like carbonyl cyanide 3-chlorophenylhydrazone (CCCP; Sigma-Aldrich Chemicals Pvt. Ltd.) on colistin resistance reversion was detected by determining the changes in minimum inhibitory concentration (MIC) of Col^R isolates. Stock solution of CCCP was prepared at 5 mg/mL and effective concentration was taken as 10 mg/L. The cultures were grown in presence and absence of CCCP with colistin in tissue culture plates and changes in MIC were calculated.²⁰ Wells containing only CCCP and culture were taken as control.

Plasmid-mediated colistin resistance. Selected known mobilized colistin resistance determinants (*mcr-1*, *mcr-2*, *mcr-3*, *mcr-4*, and *mcr-5* genes) and its variants from previous literature were targeted through multiplex PCR as per reference.²¹

Detection of mutations in *lpxA/C/D* and *pmrA/B* genes. Mutation in the genes involved in lipid A biosynthesis and point mutation in PmrA/B two-component regulatory system were detected by amplifying the *lpxA*, *lpxC*, *lpxD*, *pmrA*, and *pmrB* genes. Primers were designed in-house from flanking regions and those covering the entire gene (Table 1). Each primer was then validated with amplification and sequencing of particular gene of a reference strain (*A. baumannii* ATCC 19606; wild-type strain). Each single reaction mixture (25 µL) contained 2.5 µL Taq DNA buffer, 2 µL of dNTP, and 1 µL of each primer (10 picomole; Eurofins Scientific), 0.3 µL of Taq DNA polymerase (Genei, Bangalore, India). To maintain volume, 5 µL of template DNA (100 ng/µL) and nuclease-free water was added. Reactions were run under the following conditions for *lpxA*, *pmrA*, and *pmrB* gene amplification: initial denaturation at 95°C for 1 min, 30 cycles at 95°C for 15 sec, 48°C for 40 sec, 72°C for 30 sec, and final extension at 72°C for 7 min. For *lpxC* and *lpxD* reaction, conditions used were initial denaturation at 95°C for 3 min, 30 cycles at 95°C for 30 sec, 57°C for 45 sec, 72°C for 1 min, and final extension at 72°C for 7 min. Amplified product was identified by agarose gel electrophoresis. The amplicon obtained from all 9 Col^R isolates and the wild-type strain was sequenced by Sanger sequencing (Eze Diagon Healthcare Pvt. Ltd.). Analysis of sequencing result was performed using clustal W multiple sequence alignment tool. The mutations in the nucleotide were noted from the aligned sequences. Following this, *in vitro* nucleotide sequence translation was created on ExPASy translate tool. These translated sequences were again aligned with the help of clustal W multiple sequence alignment tool and any change in amino acid was noted.

TABLE 1. PRIMER SEQUENCES (IN-HOUSE DESIGNED) USED IN THE STUDY

S. No.	Primer pairs	Sequence (5'-3')	Target	bp
1	LpxA-F LpxA-R	ATTAGCGCACAAATTCACGCT ATGACCGAGTCTACTACACCT	<i>lpxA</i>	1,283
2	LpxC-F LpxC-R	CAATGACTTATGTCACTCAC GTGAGTGTGACATAAGTCATTG	<i>lpxC</i>	913
3	LpxD-F LpxD-R	GTGATTTGGGTCAATGGCA CGACTGTTGCCTATGACGCT	<i>lpxD</i>	1,079
4	PmrA-F PmrA-R	ATTATGATTGCCCCAAACGGT AGCGAGCCTAGAACATGACA	<i>pmrA</i>	690
5	PmrB-F PmrB-R	TACGTAATCACGCTCTTGT AGTGCTTATAAGGTTGCACT	<i>pmrB</i>	1,263

Analysis of LPS by sodium dodecyl sulfate–polyacrylamide gel electrophoresis. LPS from colistin-resistant isolates and control strain *A. baumannii* ATCC 19606 was extracted using the hot aqueous–phenol protocol described by Mofatt *et al.* with some modifications.²² The run was performed on 12% sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE). Visualization of LPS was done by silver staining.

Study of virulence factors

Biofilm formation. Biofilm quantification was performed on presterilized 96-well flat-bottom polystyrene microtiter plate as previously described with slight modification.^{23,24} Briefly, overnight grown cultures were diluted in Luria Bertani (LB) broth medium (1:100). Ten microliters of diluted suspension ($\sim 1.5 \times 10^8$ colony-forming unit [CFU]/mL) was transferred into each well containing 190 μ L of LB medium with 0.25% glucose. Following overnight incubation at 37°C, 1% crystal violet was used for biofilm quantification. Wells containing only media were taken as negative control. Biofilm formation was observed under enzyme-linked immunosorbent assay plate reader (Eli-LIMS; Transasia Bio Medical Ltd.) by measuring the optical density (OD) at 600 nm. The experiment was carried out in triplicates. Isolates were categorized as non-biofilm producer and weak/moderate/strong biofilm producers based on their OD values. OD category for non-biofilm producer and biofilm producers in each category was decided by calculating three standard deviations above the mean OD of clean microtiter plate containing only media and stained by the explained method. The categories were as follows: non-biofilm producer: $OD_{600} < 0.015$; weak-biofilm producer: $0.015 \leq OD_{600} < 0.03$; medium-biofilm producer: $0.03 \leq OD_{600} < 0.045$; and strong-biofilm producer: $0.045 \leq OD_{600}$.

Growth kinetics. Growth rate in Col^R strain was compared with Col^S by standard growth curve analysis.²⁵ Briefly, pure isolated colonies of *A. baumannii* were inoculated into 50 mL of LB broth and kept for overnight incubation at 37°C in a shaker incubator with constant shaking at 180 rpm to obtain a seed culture. Freshly prepared 50 mL of LB medium was inoculated with 10 μ L ($\sim 0.5 \times 10^8$ CFU/mL) of each of these overnight seed cultures and incubated at 37°C under constant shaking. Their cell count was estimated at time intervals of 0, 1, 2, 3, 4, 6, and 8 hr on LB agar plates. The growth rate constant and doubling time of each strain were calculated using standard formula.

Desiccation survival assay. Desiccation tolerance was tested over a period of 125 days. From overnight culture, 1 mL was taken into sterile 1 mL tube and centrifuged at 15,000 rpm for 5 min. The pellets were washed twice with sterile distilled water and subsequently resuspended in 1 mL of distilled water. From this suspension, 20 μ L was transferred onto cover slips (dimension 22 mm) and kept inside covered petriplates. These petriplates were kept within sterile plastic boxes with relative humidity inside the plastic boxes being maintained at $31\% \pm 3\%$ by the presence of a saturated salt solution of CaCl₂. Viable counts were determined at every 20 days' interval upto ~ 125 days as described by Jawad *et al.*²⁶

Typing assay. The epidemiological link between Col^R isolates was studied by repetitive extragenic palindromic PCR.²⁷ Dendrogram was prepared using BioNumerics version 7.5 software (Applied Maths, Inc., Austin, TX). Gel lanes and bands were detected; cluster analysis was done by neighbor-joining clustering method with the Jaccard similarity coefficient.

Statistical analysis

The expression of virulence factors like degree of biofilm formation, rate of growth and comparison of desiccation survival curve in resistant and susceptible isolates of *A. baumannii* was compared using statistical tool such as *t*-test for the former two virulence factors and Log-rank (Mantel-Lox) test and Gehan-Breslow-Wilcoxon test for the latter one with the help of GraphPad prism version 5 Software (La Jolla, CA).

Results

Patient details

The detail information about the patients from whom the Col^R isolates were identified revealed that majority of the patients were located in the ICUs. Colistin had been administered in only three of the nine patients (33.33%), while all the patients (100%) were administered carbapenems. Of the total, mortality was seen in four patients (44.44%). The details have been provided in Table 2.

Antimicrobial susceptibility testing

The *in vitro* screening showed non-susceptibility against majority of the antibiotics. However, different antibiogram

TABLE 2. DETAILED INFORMATION OF PATIENT WITH COLISTIN-RESISTANT ACINETOBACTER BAUMANNII INFECTIONS

Patient No. (isolate No.)	Age/sex	Location	Culture site	Type of infection	Prior colistin used	Antibiotic therapy	Mortality
P1 (Col ^R 1)	65/M	ICU 1	ETT aspirate	Acute exacerbation, septic shock	Yes	Meropenem, Linezolid, Colistin methanesulfonate	Expired
P2 (Col ^R 2)	45/M	ICU 2	ETT aspirate	VAP	No	Piperacillin/tazobactam, Imipenem, Linezolid, Ampicillin/Sulbactam	Survived
P3 (Col ^R 3)	21/M	ICU 1	ETT aspirate	Meningitis	No	Polymyxin B, Ampicillin/Sulbactam	Expired
P4 (Col ^R 4)	23/F	Bum ward	Pus	Sepsis	Yes	Piperacillin/tazobactam, Imipenem, Colistin methanesulfonate	Expired
P5 (Col ^R 5)	55/M	Surgery ward	Pus	Post of wound infection	No	ceftriaxone, Linezolid, imipenem	Survived
P6 (Col ^R 6)	25/M	ICU 2	ETT aspirate	VAP	No	Piperacillin/tazobactam, Imipenem/Cilastatin, Vancomycin, Clindamycin, polymyxin B	Survived
P7 (Col ^R 7)	40/M	Surgery ward	Pus	Post of wound infection	No	Clindamycin, polymyxin B	Survived
P8 (Col ^R 8)	40/M	ICU 1	ETT aspirate	VAP	Yes	Meropenem, Colistin methanesulfonate	Survived
P9 (Col ^R 9)	70/M	Male Surgical Ward	Tissue	Abdominal wound infection	No	Minocycline, Imipenem, Teicoplanin	Expired

Col^R, colistin resistant; ETT, endotracheal tube; F, female; ICU, intensive care unit; M, male; VAP, ventilator-associated pneumonia.

types were obtained through minimum inhibitory concentration (Table 3). The MIC for colistin ranged from 4 to ≥ 64 $\mu\text{g/mL}$ (Table 4).

Detection of carbapenemase genes

All the nine Col^R isolates carried *bla*_{OXA-23} and one or more class B carbapenemase genes. However, no class A carbapenemase gene was detected (Table 4).

Study of mechanism of colistin resistance

Greater than eightfold MIC decrease was detected in seven of nine (77.77%) Col^R isolates in the presence of CCCP as an efflux pump inhibitor. None of the *mcr* genes or its variants was detected. The results have been summarized in Table 4.

Different mutations were observed in all the nine Col^R isolates in *pmrA/B* and *lpxA/D* genes, which have been tabulated in Table 5. Isolates Col^R1, Col^R3, Col^R4, and Col^R9 showed a single-base deletion in *lpxA* gene, resulting in amino acid frameshift. The isolates Col^R1 and Col^R3 showed a single-base deletion causing truncated PmrB protein, while Col^R9 contained substitution of single nucleotide in *pmrA* gene resulting in amino acid changes. Similarly, Col^R5, Col^R6, Col^R7, and Col^R8 isolates had mutation in *lpxA* gene with a single-base deletion or insertion resulting in truncated protein. In addition, Col^R7 and Col^R8 had single base substitution in *pmrA* gene with amino acid change. Only one isolate Col^R2 was found with insertion mutation in *lpxD* gene, resulting in truncated protein. However, no mutation was reported in *lpxC* gene.

On SDS-PAGE, none of the isolates having mutation in *lpxA* gene showed the presence of LPS as detected by the absence of a band at 8 kDa in comparison to *A. baumannii* ATCC 19606 (Supplementary Fig. S1).

Study of virulence factors

Degree of biofilm formation revealed that both the resistant (9, 100%) and susceptible (6, 67%) isolates were significantly strong biofilm producers ($p=0.04$) (Fig. 1). The mean OD₆₀₀ for Col^R and Col^S was found to be 0.16 and 0.08, respectively, both being in the “strong biofilm” category. Colonies of Col^R isolates were smaller in size in comparison to Col^S isolates. Col^S isolates were able to survive desiccation better (more than 125 days) than the Col^R isolates (125 days) (Fig. 2). The proportions of Col^S and Col^R isolates that survived desiccation after 60, 100, and 125 days were 8/8, 6/4, and 6/1, respectively. However, the difference in survival proportions was not significant ($p=0.18$). The growth rate (μ) and doubling time (g) of Col^R isolates were found to be significantly less and more, respectively, than the Col^S isolates ($p=0.03$). For all the Col^R and Col^S isolates, the mean μ and g calculated during the exponential phase were $\mu=1.06\text{ hr}^{-1}$, $g=39\text{ min}$ and $\mu=1.48\text{ hr}^{-1}$, $g=28\text{ min}$, respectively (Fig. 3).

Typing assay

The dendrogram (Fig. 4) revealed similarity between Col^R1 and Col^R3 isolates that were isolated from two different patients in the same ICU at the same time. Another clone comprising two isolates with more than 60% similarity was also seen in the male surgical ward.

TABLE 3. RESISTANCE PROFILE OF COLISTIN-RESISTANT *ACINETOBACTER BAUMANNII* ISOLATES

Isolate No.	Antimicrobial susceptibility																	
	Disc diffusion										MIC values ($\mu\text{g/mL}$)							
	LVX	CIP	SXT	AMI	GEN	CPM	CAZ	CRO	PTZ	AMS	IPM	MEM	DOR	PB	CL	TGC	MIN	
Col ^R 1	R	R	R	R	R	R	R	R	R	I	>256	>128	64	>64	4	2	1	
Col ^R 2	R	R	R	R	R	R	R	R	R	R	128	>128	64	64	64	2	2	
Col ^R 3	R	R	R	R	R	R	R	R	R	I	>256	64	>128	>64	64	8	8	
Col ^R 4	R	R	R	R	R	R	R	R	R	R	128	64	32	64	64	16	8	
Col ^R 5	R	R	R	R	R	R	R	R	R	R	128	128	64	32	64	8	8	
Col ^R 6	R	R	R	R	R	R	R	R	R	R	128	64	>128	32	64	4	2	
Col ^R 7	R	R	R	R	R	R	R	R	R	I	256	>128	>128	4	32	8	32	
Col ^R 8	R	R	R	R	R	R	R	R	R	R	128	128	>128	4	32	4	8	
Col ^R 9	R	R	R	R	R	R	R	R	R	R	128	64	32	4	64	8	32	

AMS, ampicillin/sulbactam; CAZ, ceftazidime; CIP, ciprofloxacin; CL, colistin; CPM, cefepime; CRO, ceftriaxone; DOR, doripenem; GEN, gentamicin; I, intermediate; IPM, imipenem; LVX, levofloxacin; MEM, meropenem; MIN, minocycline; PB, polymyxin B; PTZ, piperacillin/tazobactam; R, resistant; SXT, trimethoprim/sulfamethoxazole; TGC, tigecycline.

Discussion

A. baumannii has been recognized as one of the important nosocomial pathogens with respect to MDR. Among antibiotics, colistin and tigecycline are the last-resort drugs for treating serious infections caused by these resistant organisms. This study showed resistance rate of 2.5% among *A. baumannii* isolates against colistin in a tertiary care hospital in India. The SENTRY surveillance (2001–2011) encompassing various centers for global representation has reported colistin resistance in *A. baumannii* in the range of 0.9–3.3%, with reports from Spain, Korea, Taiwan, and Bulgaria showing alarming situations.⁹ There have been scarce data from India on this aspect with one report of eight resistant isolates (3.5%) detected from complicated urinary tract infection in patients with prior intravenous colistin administration and another single resistant isolate from tracheal secretions of a patient in septic shock from the ICU with prior colistin therapy.^{12,28} However, both these studies had employed E-test method for detection of colistin MIC.

In this study, only three of the nine patients (33.33%) were administered colistin before the isolation of resistant isolates. A recent study, however, reported that colistin resistance had emerged exclusively from use of therapeutic colistin in 19 of the 20 patients infected with colistin-

resistant *A. baumannii*.¹⁰ Previous use of colistin therapy has been considered a risk factor for the development of colistin resistance.⁹ Polymyxin resistance usually occurs under selective pressure through mutations in certain determinants as a response to it.⁴ Therefore, it is hypothesized that in presence of polymyxins, these resistant isolates occur randomly and in absence of antibiotics, these isolates might not persist owing to high fitness cost of the organism. A similar study in this context had demonstrated that susceptible isolates out-competed resistant ones upon withdrawal of colistin treatment.²⁹ However, in this study, majority of resistant isolates emerged without colistin therapy. In this regard, studies have suggested that prior exposure to carbapenems in low-endemicity carbapenem resistance settings is an independent risk factor for development of colistin resistance in Enterobacteriaceae.³⁰ In high-endemicity setups like ours, emergence of colistin resistance has been in parallel with the dissemination of carbapenemases.^{14,31} Carbapenemases were detected in all the study isolates. Therefore, with prior carbapenem therapy in all the patients, colistin resistance could have emerged under their selective pressure. However, this is just an assumption and requires further validation. Mortality was nearly 50% in this study, although it could not be attributed to any specific cause as patients were

TABLE 4. MINIMUM INHIBITORY CONCENTRATION VALUE OF COLISTIN, EFFECT OF CARBONYL CYANIDE 3-CHLOROPHENYLHYDRAZONE ON COLISTIN MIC, AND PRESENCE OF CARBAPENEMASE AND *mcr* GENES IN COLISTIN-RESISTANT *ACINETOBACTER BAUMANNII* ISOLATES

A. baumannii isolate No.	MIC of colistin ($\mu\text{g/mL}$)	MIC of colistin+CCCP ($\mu\text{g/mL}$)	Carbapenemase genes			mcr-genes
			Class D	Class A	Class B	
Col ^R 1	4	0.5	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{NDM} , <i>bla</i> _{IMP}	—
Col ^R 2	64	0.5	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{NDM} , <i>bla</i> _{IMP}	—
Col ^R 3	64	8	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{NDM} , <i>bla</i> _{IMP}	—
Col ^R 4	64	32	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{NDM} , <i>bla</i> _{IMP}	—
Col ^R 5	64	0.5	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{NDM} , <i>bla</i> _{IMP}	—
Col ^R 6	64	1	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{IMP}	—
Col ^R 7	32	2	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{IMP}	—
Col ^R 8	32	16	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{IMP}	—
Col ^R 9	64	4	<i>bla</i> _{OXA-23}	—	<i>bla</i> _{NDM}	—

CCCP, carbonyl cyanide 3-chlorophenylhydrazone.

TABLE 5. MUTATIONS IDENTIFIED IN *LPXA/C/D* AND *PMRA/B* GENES OF COLISTIN-RESISTANT *ACINETOBACTER BAUMANNII* ISOLATES

Isolate	Gene	Nucleotide change	Effect
Col ^R 1	<i>pmrB</i>	T deletion at nt 225	Truncation after A 75
	<i>lpxA</i>	A deletion at nt 44	Frameshift after E 15
Col ^R 2	<i>lpxD</i>	T insertion at nt 291	Truncation after S 98
Col ^R 3	<i>pmrB</i>	T deletion at nt 232	Truncation after R 77
	<i>lpxA</i>	A deletion at nt 44	Frameshift after E 15
Col ^R 4	<i>lpxA</i>	A deletion at nt 44	Frameshift after E 15
Col ^R 5	<i>lpxA</i>	T deletion at nt 594	Truncation after T 198
Col ^R 6	<i>lpxA</i>	T insertion at nt 621	Truncation after L 207
Col ^R 7	<i>pmrA</i>	G→A substitution at nt 226	Amino acid change A 76 T
	<i>lpxA</i>	T deletion at nt 594	Truncation after T 198
Col ^R 8	<i>pmrA</i>	G→A substitution at nt 43	Amino acid change D 14 N
	<i>lpxA</i>	T insertion at nt 621	Truncation after L 207
Col ^R 9	<i>pmrA</i>	G→A substitution at nt 43	Amino acid change D 14 N
	<i>LpxA</i>	A deletion at nt 44	Frameshift after E 15

critically ill, which could have affected their mortality. There has been no consensus on attributable mortality due to *Acinetobacter* infections, owing to the difficulty in ascertaining their colonizer/pathogen status.³² In addition, outcomes in patients with colistin resistance *A. baumannii* might depend on the various mechanisms of resistance specific for the strains.¹³

Mechanism of colistin resistance in *A. baumannii* has not yet been clearly elucidated, despite several studies.^{10,33} Decreased susceptibility has been reported in earlier studies, owing to mutation as a sole mechanism. Mutation in *pmrA/B* or in *lpxA/C/D* genes is the principle reason for colistin resistance.^{8,34} Multiple mutations in *pmrA/B* and *lpxA/D* genes at novel sites were detected among these resistant isolates. Most of the studies have reported *pmrB* mutations.^{25,33,34} This study detected *pmrA* mutations at A76T in one isolate, D14N in two isolates, and *pmrB* mutations at A75 and R77 in two isolates, respectively. In studies elsewhere, mutations in *pmrA* gene have been reported in majority or all the polymyxin-resistant *A. baumannii* isolates at different sites like P102H and D82G, and S119T.^{3,11} The

study by Leite *et al.* had also mentioned that the expression of *pmrA* in these isolates occurred even in absence of colistin, thus indicating that elevated expression of *pmrA* was constitutive and not inductive.¹¹ Apart from mutations, contributions of efflux pump and acquired resistance due to horizontal gene transfer of *mcr*-genes have also been advocated for colistin-resistant isolates. The role of efflux pump inhibitors like CCCP in reversing colistin resistance through decrease in MIC values was seen in majority of colistin-resistant isolates (77.77%), and has also been reported in some of the recent studies.^{35,36} From these observations, it could be predicted that LPS modification/loss due to mutations or an efflux pump system could be the commonest mechanism of colistin resistance in these isolates. Efflux pump inhibitors like CCCP might have restored the membrane properties as suggested by the above studies. Threat of wide dissemination of *mcr* genes in multidrug-resistant organisms, including *A. baumannii*, has been proposed based on studies on cloned constructs of *A. baumannii*.³⁷ There have been limited reports on *mcr* gene in *A. baumannii*, as was also concluded from this study.³⁸

A. baumannii has tremendous ability to colonize both viable and damaged tissues.⁴ Formation of biofilms is a

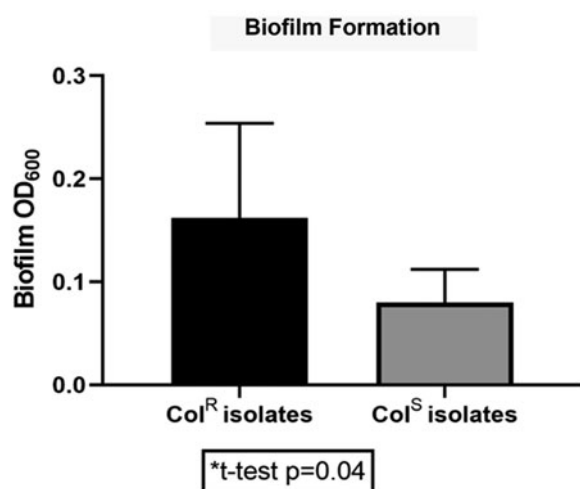


FIG. 1. Comparison of degree of biofilm formation in Col^R and Col^S *Acinetobacter baumannii* isolates. Col^R, colistin resistant; Col^S, colistin susceptible.

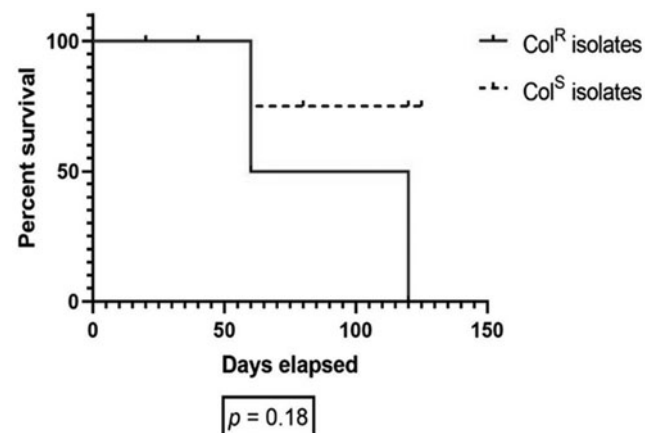


FIG. 2. Comparison of desiccation survival in Col^R and Col^S *Acinetobacter baumannii* isolates.

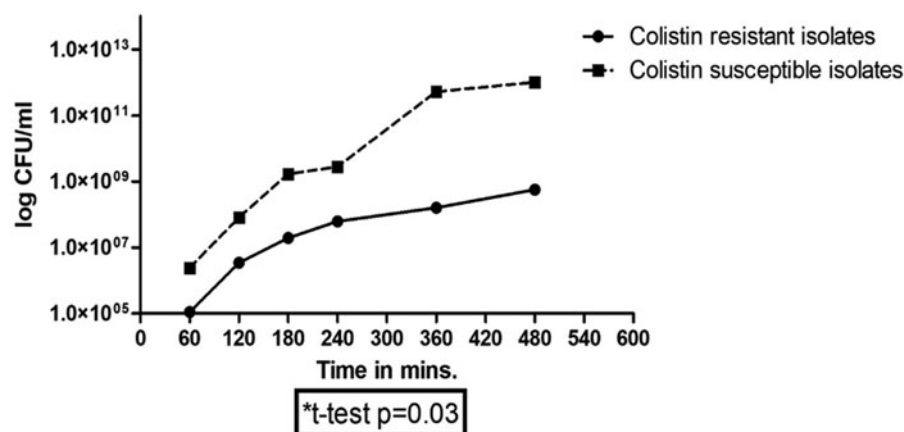


FIG. 3. Comparison of *in vitro* growth rate (mean value) of the Col^R and Col^S *Acinetobacter baumannii* isolates.

potent virulence attribute of *A. baumannii* along with resistance to desiccation to survive and persist in the hospital environment. It has been hypothesized that modulation because of mutation or overburden of bacterial genome due to many antibiotic-resistant determinants can compromise these biological processes of bacteria.^{13,25} Consequently, colistin resistance mechanisms might affect the fitness cost of the organism often manifested as reduced biofilm formation, growth, desiccation survival, and motility.^{36,37,39–41} This study revealed no difference in biofilm forming ability of the resistant and susceptible isolates, which could have been due to isolation of these isolates from hospitalized patients. Similarly, both the group of isolates survived desiccation without any significant difference, but the resistant isolates were significantly slow growers compared to the susceptible ones.

Studies have shown that fitness and virulence of *A. baumannii* decrease with acquisition of colistin resistance, while in treatment with colistin.⁴² In one of the studies, no significant difference in growth rate was observed when both colistin-resistant and colistin-susceptible isolates were grown separately, but growth of resistant isolates was reduced when grown competitively.¹³ On the contrary, there have been

studies revealing loss of virulence and fitness of colistin-resistant *A. baumannii* isolates only initially, to be followed by regaining of both under oxidative stress.⁴³ Different genetic backgrounds of the isolates also affect their virulence and fitness. For colistin-resistant *A. baumannii* isolates collected from clinical sources, there have been inconsistent reports of *in vivo* fitness cost and impaired virulence compared to laboratory-induced mutants. Thus, the clinical background and various complex factors also influence the association of fitness cost with virulence factors.¹³ In this context, all the resistant isolates in this study did not show any significant compensation in their virulence, despite their colistin-resistant status. Hence, we concluded that there was no reduction in fitness of these resistant isolates.

We studied the current status of colistin resistance in *A. baumannii* from a single center in India, which might not represent the actual situation at large. We could not study the attributable mortality of these isolates as we relied only on clinician's suspicion of these isolates as pathogens. The role of carbapenems in emergence of these isolates could not be studied. Nonetheless, this study reveals the varying mechanisms of resistance adopted by the isolates in a setup of intense carbapenem use. This is the first in-depth study to

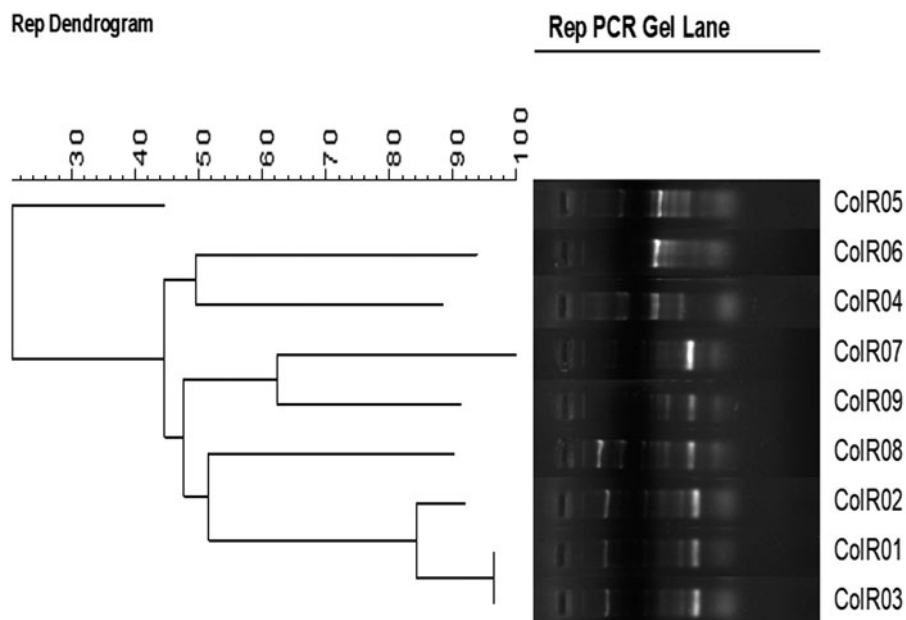


FIG. 4. Dendrogram of the Col^R *Acinetobacter baumannii* isolates by rep-PCR. rep-PCR, repetitive extragenic palindromic polymerase chain reaction.

the best of our knowledge, encompassing the resistance mechanisms and changes in fitness of colistin-resistant *A. baumannii* isolates from India.

Conclusion

Colistin resistance rate of 2.5% in *A. baumannii* was studied from a tertiary care center in India, with mutations at novel sites in *pmrA/B* and *lpxA/D* genes as major resistance mechanism along with reversion of resistance with CCCP. Majority of these isolates were collected from patients without prior colistin therapy. The resistant isolates demonstrated no reduction in fitness with respect to virulence factors like biofilm formation and desiccation survival, but were significant slow growers than the susceptible ones.

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Disclosure Statement

No competing financial interests exist.

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Supplementary Material

Supplementary Figure S1

References

- World Health Organization. 2013. Global Priority List of Antibiotic-Resistant Bacteria to Guide Research, Discovery, and Development of New Antibiotics. WHO Press, Geneva, pp. 1–7.
- Liu, Y.Y., Y. Wang, T.R. Walsh, L.X. Yi, R. Zhang, J. Spencer, Y. Doi, G. Tian, B. Dong, X. Huang, L.F. Yu, D. Gu, H. Ren, X. Chen, L. Lv, D. He, H. Zhou, Z. Liang, J.H. Liu, and J. Shen. 2016. Emergence of plasmid-mediated colistin resistance mechanism MCR-1 in animals and human beings in China: a microbiological and molecular biological study. *Lancet Infect. Dis.* 16:161–168.
- Ahmed, S.S., E. Alp, J. Hopman, and A. Voss. 2016. Global epidemiology on colistin resistant *Acinetobacter baumannii*. *J. Infect. Dis. Ther.* 4:1–5.
- Lean, S.S., Z. Suhaili, S. Ismail, Z. Jusoh, C.C. Yeo, and K.L. Thong. 2014. Prevalence and genetic characterization of carbapenem- and polymyxin-resistant *Acinetobacter baumannii* isolated from a tertiary hospital in Terengganu, Malaysia. *ISRN Microbiol.* 2014:953417.
- Boral, B., Ö. Unaldi, A. Ergin, R. Drumaz, and O.K. Eser; Acinetobacter Study Group. 2019. A prospective multi-center study on the evaluation of antimicrobial resistance and molecular epidemiology of multidrug-resistant *Acinetobacter baumannii* infections in intensive care units with clinical and environmental features. *Ann. Clin. Microbiol. Antimicrob.* 18:19.
- Wand, M.E., L.J. Bock, L.C. Bonney, and J.M. Sutton. 2015. Retention of virulence following adaptation to colistin in *Acinetobacter baumannii* reflects the mechanism of resistance. *J. Antimicrob. Chemother.* 70:2209–2216.
- Boinett, C.J., A.K. Cain, J. Hawkey, N.T. Do Hoang, N.N.T. Khanh, D.P. Thanh, J. Dordel, J.I. Campbell, N.P.H. Lan, M. Mayho, G.C. Langridge, J. Hadfield, N.V.V. Chau, G.E. Thwaites, J. Parkhill, N.R. Thomson, K.E. Holt, and S. Baker. 2019. Clinical and laboratory-induced colistin-resistance mechanisms in *Acinetobacter baumannii*. *Microb. Genom.* 5:e000246.
- Vila-Farrés, X., M. Ferrer-Navarro, A.E. Callarisa, S. Martí, P. Espinal, S. Gupta, J.M. Rolain, E. Giralt, and J. Vila. 2015. Loss of LPS is involved in the virulence and resistance to colistin of colistin-resistant *Acinetobacter nosocomialis* mutants selected in vitro. *J. Antimicrob. Chemother.* 70:2981–2986.
- Cai, Y., D. Chai, R. Wang, B. Liang, and N. Bai. 2012. Colistin resistance of *Acinetobacter baumannii*: clinical reports, mechanisms and antimicrobial strategies. *J. Antimicrob. Chemother.* 67:1607–1615.
- Qureshi, Z.A., L.E. Hittle, J.A. O'Hara, J.I. Rivera, A. Syed, R.K. Shields, A.W. Pasculle, R.K. Ernst, and Y. Doi. 2015. Colistin-resistant *Acinetobacter baumannii*: beyond carbapenem resistance. *Clin. Infect. Dis.* 60:1295–1303.
- Leite, G.C., R.A. Stabler, P. Neves, L.V. Perdigão Neto, R.C. Ruedas Martins, C. Rizek, F. Rossi, A.S. Levin, and S.F. Costa. 2019. Genetic and virulence characterization of colistin-resistant and colistin-sensitive *A. baumannii* clinical isolates. *Diagn. Microbiol. Infect. Dis.* 95:99–101.
- Sinha, S., S. Sahu, J. Pati, B. Ray, and S.K. Pattnaik. 2019. Retrospective analysis of colistin-resistant bacteria in a tertiary care centre in India. *Indian J. Med. Res.* 149:418–422.
- Da Silva, G., and S. Domingues. 2017. Interplay between colistin resistance, virulence and fitness in *Acinetobacter baumannii*. *Antibiotics (Basel)* 6:28.
- Banerjee, T., A. Mishra, A. Das, S. Sharma, H. Barman, and G. Yadav. 2018. High prevalence and endemicity of multidrug resistant *Acinetobacter* spp. in intensive care unit of a Tertiary Care Hospital, Varanasi, India. *J. Pathog.* 2018:9129083.
- Chen, T.L., Y.T. Lee, S.C. Kuo, S.P. Yang, C.P. Fung, and S.D. Lee. 2014. Rapid identification of *Acinetobacter baumannii*, *Acinetobacter nosocomialis* and *Acinetobacter pittii* with a multiplex PCR assay. *J. Med. Microbiol.* 63(Pt 9):1154–1159.
- Marchaim, D., J.M. Pogue, O. Tzuman, K. Hayakawa, P.R. Lephart, H. Salimnia, T. Painter, M.J. Zervos, L.E. Johnson, M.B. Perri, P. Hartman, R.V. Thyagarajan, S. Major, M. Goodell, M.G. Fakih, L.L. Washer, D.W. Newton, A.N. Malani, J.M. Wholehan, L. Mody, and K.S. Kaye. 2014. Major variation in MICs of tigecycline in Gram-negative bacilli as a function of testing method. *J. Clin. Microbiol.* 52:1617–1621.
- Clinical and Laboratory Standards Institute. 2019. Performance Standards for Antimicrobial Susceptibility Testing: 29th ed., CLSI Supplement M100. CLSI, Wayne, PA.
- Poirel, L., T.R. Walsh, V. Cuvillier, and P. Nordmann. 2011. Multiplex PCR for detection of acquired carbapenemase genes. *Diagn. Microbiol. Infect. Dis.* 70:119–123.
- Turton, J.F., N. Woodford, J. Glover, S. Yarde, M.E. Kaufmann, and Pitt TL. 2006. Identification of *Acinetobacter baumannii* by detection of the bla OXA-51-like carbapenemase gene intrinsic to this species. *J. Clin. Microbiol.* 44:2974–2976.

20. Baron, S.A., and J.M. Rolain. 2018. Efflux pump inhibitor CCCP to rescue colistin susceptibility in mcr-1 plasmid-mediated colistin-resistant strains and Gram-negative bacteria. *J. Antimicrob. Chemother.* 73:1862–1871.
21. Rebelo, A.R., V. Bortolaia, J.S. Kjeldgaard, S.K. Pedersen, P. Leekitcharoenphon, I.M. Hansen, B. Guerra, B. Malorny, M. Borowiak, J.A. Hammerl, A. Battisti, A. Franco, P. Alba, A. Perrin-Guyomard, S.A. Granier, C. De Frutos Escobar, S. Malhotra-Kumar, L. Villa, A. Carattoli, and R.S. Hendriksen. 2018. Multiplex PCR for detection of plasmid-mediated colistin resistance determinants, mcr-1, mcr-2, mcr-3, mcr-4 and mcr-5 for surveillance purposes. *Euro Surveill.* 23:17-00672.
22. Moffatt, J.H., M. Harper, and P. Harrison. 2010. Colistin resistance in *Acinetobacter baumannii* is mediated by complete loss of lipopolysaccharide production. *Antimicrob. Agents Chemother.* 54:4971–4977.
23. Greene, C., G. Vadlamudi, D. Newton, B. Foxman, and C. Xi. 2016. The influence of biofilm formation and multidrug resistance on environmental survival of clinical and environmental isolates of *Acinetobacter baumannii*. *Am. J. Infect. Control* 44:e65–e71.
24. Badave, G.K., and D. Kulkarni. 2015. Biofilm producing multidrug resistant *Acinetobacter baumannii*: an emerging challenge. *J. Clin. Diagn. Res.* 9:DC08–DC10.
25. Beceiro, A., A. Moreno, N. Fernández, J.A. Vallejo, J. Aranda, B. Adler, M. Harper, J.D. Boyce, and G. Bou. 2014. Biological cost of different mechanisms of colistin resistance and their impact on virulence in *Acinetobacter baumannii*. *Antimicrob. Agents Chemother.* 58:518–526.
26. Jawad, A., H. Seifert, A.M. Snelling, J. Heritage, and P.M. Hawkey. 1998. Survival of *Acinetobacter baumannii* on dry surfaces: comparison of outbreak and sporadic isolates. *J. Clin. Microbiol.* 36:1938–1941.
27. Fitzpatrick, M.A., E.A. Ozer, and A.R. Hauser. 2016. Utility of whole-genome sequencing in characterizing *Acinetobacter* epidemiology and analyzing hospital outbreaks. *J. Clin. Microbiol.* 54:593–612.
28. Taneja, N., G. Singh, M. Singh, and M. Sharma. 2011. Emergence of tigecycline & colistin resistant *Acinetobacter baumannii* in patients with complicated urinary tract infections in north India. *Indian J. Med Res.* 133:681–684.
29. Snitkin, E.S., A.M. Zelazny, J. Gupta, NISC Comparative Sequencing Program, T.N. Palmore, P.R. Murray, and J.A. Segre. 2013. Genomic insights into the fate of colistin resistance and *Acinetobacter baumannii* during patient treatment. *Genome Res.* 23:1155–1162.
30. Büchler, A.C., C. Gehringer, A.F. Widmer, A. Egli, and S.T. Sutter. 2018. Risk factors for colistin-resistant enterobacteriaceae in a low-endemicity setting for carbapenem resistance—a matched case–control study. *Euro Surveill.* 23:1700777.
31. Monaco, M., T. Giani, M. Raffone, F. Arena, A. Garcia-Fernandez, S. Pollini, Network EuSCAPE-Italy, H. Grundmann, A. Pantosti, and G.M. Rossolini. 2014. Colistin resistance superimposed to endemic carbapenem-resistant *Klebsiella pneumoniae*: a rapidly evolving problem in Italy, November 2013 to April 2014. *Euro Surveill.* 19:20939.
32. Gordon, N.C., and D.W. Wareham. 2010. Multidrug-resistant *Acinetobacter baumannii*: mechanisms of virulence and resistance. *Int. J. Antimicrob. Agents* 35:219–226.
33. Zhang, W., B. Aurosree, B. Gopalakrishnan, J.M. Balada-Llasat, V. Pancholi, and P. Pancholi. 2017. The role of LpxA/C/D and pmrA/B gene systems in colistin-resistant clinical strains of *Acinetobacter baumannii*. *Front. Lab. Med.* 1:86–91.
34. Beceiro, A., E. Llobet, J. Aranda, J.A. Bengoechea, M. Doumith, M. Hornsey, H. Dhanji, H. Chart, G. Bou, D.M. Livermore, and N. Woodford. 2011. Phosphoethanolamine modification of lipid A in colistin-resistant variants of *Acinetobacter baumannii* mediated by the pmrAB two-component regulatory system. *Antimicrob. Agents Chemother.* 55:3370–3379.
35. Park, Y.K., and K.S. Ko. 2015. Effect of carbonyl cyanide 3-chlorophenylhydrazone (CCCP) on killing *Acinetobacter baumannii* by colistin. *J. Microbiol.* 53:53–59.
36. Ni, W., Y. Li, J. Guanet, J. Zhao, J. Cui, R. Wang, and Y. Liu. 2016. Effects of efflux pump inhibitors on colistin resistance in multidrug-resistant gram-negative bacteria. *Antimicrob. Agents Chemother.* 60:3215–3218.
37. Liu, Y.Y., C.E. Chandler, L.M. Leung, C.L. McElheny, R.T. Mettus, R.M.Q. Shanks, J.H. Liu, D.R. Goodlett, R.K. Ernst, and Y. Doi. 2017. Structural modification of lipopolysaccharide conferred by mcr-1 in gram-negative ESKAPE pathogens. *Antimicrob. Agents Chemother.* 61:e00580-17.
38. Ma, F., C. Shen, X. Zheng, Y. Liu, H. Chen, L. Zhong, Y. Liang, K. Liao, Y. Xia, G.B. Tian, and Y. Yang. 2019. Identification of a novel plasmid carrying mcr-4.3 in an *Acinetobacter baumannii* strain in China. *Antimicrob. Agents Chemother.* 63:e00133-19.
39. Carretero-Ledesma, M., M. García-Quintanilla, R. Martín-Peña, M.R. Pulido, J. Pachón, and M.J. McConnell. 2018. Phenotypic changes associated with colistin resistance due to lipopolysaccharide loss in *Acinetobacter baumannii*. *Virulence* 9:930–942.
40. Durante-Mangoni, E., M. Del Franco, R. Andini, M. Bernardo, M. Giannouli, and R. Zarrilli. 2015. Emergence of colistin resistance without loss of fitness and virulence after prolonged colistin administration in a patient with extensively drug-resistant *Acinetobacter baumannii*. *Diagn. Microbiol. Infect. Dis.* 82:222–226.
41. Pournaras, S., A. Poulou, K. Dafopoulou, Y.N. Chabane, I. Kristo, D. Makris, J. Hardouin, P. Cosette, A. Tsakris, and E. Dé. 2014. Growth retardation, reduced invasiveness, and impaired colistin-mediated cell death associated with colistin resistance development in *Acinetobacter baumannii*. *Antimicrob. Agents Chemother.* 58:828–832.
42. López-Rojas, R., M.J. McConnell, M.E. Jiménez-Mejías, J. Domínguez-Herrera, F. Fernández-Cuenca, and J. Pachón. 2013. Colistin resistance in a clinical *Acinetobacter baumannii* strain appearing after colistin treatment: effect on virulence and bacterial fitness. *Antimicrob. Agents Chemother.* 57:4587–4589.
43. Jones, C.L., S.S. Singh, Y. Alamneh, L.G. Casella, R.K. Ernst, E.P. Lesho, P.E. Waterman, and D.V. Zurawski. 2017. In vivo fitness adaptations of colistin-resistant *Acinetobacter baumannii* isolates to oxidative stress. *Antimicrob. Agents Chemother.* 61:e00598-16.

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